week 5

Chain rule (continued)

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

what if there are more than 2 functions

$$\frac{d}{dx} f(g(h(x))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Ex $\frac{d}{dx} f(2f(3f(x))) = f'(2f(3f(x))) \cdot 2f'(3f(x)) \cdot 3f'(x)$

wow that's simple

Ex

$$g(x) = \left[f\left(\frac{2}{x}\right)\right]^3$$

$$g'(x) = 3\left[f\left(\frac{2}{x}\right)\right]^2 \cdot f'\left(\frac{2}{x}\right) \cdot \left(-\frac{2}{x^2}\right)$$

Trig - Diff-erivitives

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Proof later

can be used in calculations

Ex $\lim_{x \to 0} \frac{\sin 3x}{\sin 7x} = \lim_{x \to 0} \frac{\sin 3x}{3x} \cdot \frac{7x}{\sin 7x}$

$\lim_{x \to 0} \frac{\sin 3x}{3x} = 1$, $\lim_{x \to 0} \frac{7x}{\sin 7x} = 1$

$\Rightarrow$ lim is $\frac{3}{7}$

Derrivatives

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos^2 x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x \quad \text{(exercise)}$$

$$\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = -\frac{-\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \cdot \tan x$$

Note: the trig function d.n formulas assume the argument is in radians

if x is in degrees what is the angle in radians

$$x \text{ in degrees} \longrightarrow \frac{\pi x}{180} \text{ radians}$$

if x is in degrees,

$$y = \sin\left(\frac{\pi x}{180}\right) \Rightarrow y' = \frac{\pi}{180} \cdot \cos\frac{\pi x}{180}$$

$$y'\Big|_{x=45} = \frac{\pi}{180} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360}$$

$$y_0 = y(45^\circ) = \sin(45) = \frac{\sqrt{2}}{2}$$

Equation of tangent line:

$$y - \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360}(x - 45)$$

$$\lim_{x \to 0} x^2 \csc x \cot x$$

$$= \lim_{x \to 0} x^2 \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= \lim_{x \to 0} \left(\frac{x}{\sin x}\right)^2 \cos x$$

($\frac{x}{\sin x} \to 1$, $\cos x \to 1$)

$\Rightarrow$ lim is 1.

$y = f(x)$ differentiable

$y' = f'(x) = \frac{dx}{dy}$ first derivative

Which is a function itself

if f' is differentiable:

$$\frac{d}{dx} \cdot \frac{dy}{dx} = (f'(x))' = f''(x) = \frac{d^2y}{dx^2}$$

repeat: $y^{(n)}(x) = \frac{d^n y}{dx^n} = f^{(n)}(x)$

n^{th} order derivative of $f(x)$

Ex

$f(x) = 3x^3 - 7x^2 + x$

$f'(x) = 9x^2 - 14x + 1$

$f''(x) = 18x - 14$

$f^{(3)}(x) = 18$

$f^{(4)}(x) = 0$

any dirivitive beyond that is zero

Ex $y = f(x) = \frac{1}{x-3}$

$$f'(x) = -\frac{1}{(x-3)^2}$$

$$f''(x) = \frac{2}{(x-3)^3}$$

$$f^{(3)}(x) = -\frac{2 \cdot 3}{(x-3)^4}$$

$$f^{(4)}(x) = \frac{2 \cdot 3 \cdot 4}{(x-3)^5}$$

$1 \cdot 2 \cdot 3 \cdot 4 = 4!$

$1 \cdot 2 \cdot 3 \cdots n = n!$

$$f^{(n)} = \frac{(-1)^n \cdot n!}{(x-3)^{n+1}}$$

Valid for any integer

Ex

$f(x) = \sin x$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$f^{(3)}(x) = -\cos x$

$f^{(4)}(x) = \sin x$

$n = 4k$ (k integer) $\Rightarrow$ $f^{(n)}(x) = \sin x$

$n = 4k+1$ $\quad f^{(n)}(x) = \cos x$

$n = 4k+2$ $\quad f^{(n)}(x) = -\sin x$

$n = 4k+3$ $\quad f^{(n)}(x) = -\cos x$

Ex

$y = f(x) = \sqrt{1-3x}$ Calculate f', f''

$$f'(x) = -\frac{3}{2\sqrt{1-3x}} = -\frac{3}{2}(1-3x)^{-\frac{1}{2}}$$

$$f''(x) = \frac{3}{4}(1-3x)^{-\frac{3}{2}} \cdot (-3) = -\frac{9}{4}(1-3x)^{-\frac{3}{2}} \;//$$

Exercise: Calculate $f^{(3)}(x)$

Using derivative and differentials

$y = f(x) \qquad y' = \frac{dx}{dy} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$

$dy, dx \rightarrow$ differentials

they are Δy and Δx as they $\rightarrow 0$

$$dy = y' dx \Rightarrow \boxed{\Delta y \approx y' \cdot \Delta x}$$

Can be used to approximate small changes in the value of a function

Ex $f(x) = \sqrt{3x+1}$

Estimated the change in f as x increase from $a=1$ to $x=1.08$

$\Delta y \approx f'(a) \cdot \Delta x$

$f'(x) = \frac{3}{2\sqrt{3x+1}} \qquad f'(1) = \frac{3}{2 \cdot 2} = \frac{3}{4}$

$\Delta y \simeq \frac{3}{4} \cdot 0.08 = 0.06$

Exact value: $f(1.08) = \sqrt{3 \cdot 1.08 + 1} = 2.059126$

$f(1) = \sqrt{4} = 2$

$\Delta y_{exact} = f(1.08) - f(1) = 0.059126$

Example ★ $y = f(x) = \sqrt{x}$

if x undergoes a negative change of 2% what would be the negative change in y? $\frac{\Delta x}{x}$, $\frac{\Delta y}{y}$ negative change

$\Delta y \approx f'(a) \cdot \Delta x = f'(x) \cdot \Delta x$

$f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow \Delta y \simeq \frac{\Delta x}{2\sqrt{x}}$

$\frac{\Delta y}{y} = \frac{\Delta x}{2\sqrt{x} \cdot \sqrt{x}} = \frac{1}{2}\frac{\Delta x}{x}$ $(x > 0)$

if $\frac{\Delta x}{x} = 0.02 \Rightarrow \frac{\Delta y}{y} = 0.01$ or 1%

Example what is the rate of change of a side of a